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SMART RETAIL: A DATA-DRIVEN DECISION SUPPORT SYSTEM FOR SUSTAINABLE RETAIL OPERATIONS

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0405 – Fundamentals of Data Science

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

Mythrea Shree Jayaprakash S (192424089)

Under the Supervision of

Dr. Krishnaraj Mony

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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Chennai-602105

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**SmartRetail: A Data-Driven Decision Support System for Sustainable Retail Operations**” has been carried out by **Mythrea Shree Jayaprakash (Reg. No. 192424089)** under the supervision of Dr. Krishnaraj Mony and is submitted in partial fulfilment of the requirements for the current semester of the **B.TECH Artificial Intelligence and Data Science** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.T.P Anithaashri
Professor & HOD

Dept of Cloud Computing / CSE
Saveetha School of Engineering

COURSE FACULTY

Dr. Krishnaraj Mony
Associate Professor

Predictive Engineering/ CSE
Saveetha School of Engineering

Submitted for the Capstone Project Work Viva-Voce held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

DECLARATION

I, **Mythrea Shree Jayaprakash (Reg. No. 192424402)** of the **Department of Artificial Intelligence and Data Science**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **“SmartRetail: A DataDriven Decision Support System for Sustainable Retail Operations”** is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 08/09/2026

Signature of the Student with Name

Mythrea Shree Jayaprakash (192424089)

Chennai-602105

Individual Contribution Statement

(Not applicable – this Capstone Project was carried out individually)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Mythrea shree jayaprakash (192424)	Sole author – responsible for all stages of the project	Data pipeline, ML models, dashboard & AI assistant	Model evaluation, dashboard testing, validation	All chapters	25%
Total					25%

ABSTRACT

Modern retail enterprises generate large volumes of transactional data every day across sales channels, inventory systems, and supply chains, yet many still depend on manual spreadsheet reporting to interpret this information. Such retrospective, descriptive methods struggle to reveal the correlations between discounting, shipping performance, and profitability, and offer no forward-looking capability to anticipate demand or prevent unsustainable stock accumulation. This creates a clear engineering need for an integrated, data-driven decision support system that can transform raw retail records into predictive, actionable intelligence.

This project presents SmartRetail, a Python-based data-driven decision support system designed to analyse, forecast, and optimise retail operations while promoting sustainable inventory practices aligned with UN Sustainable Development Goal 12 (Responsible Consumption and Production). The system follows a modular pipeline comprising automated data acquisition and preprocessing (missing-value imputation, duplicate removal, interquartilerange outlier filtering, and feature engineering of shipping duration and profit margin); multidimensional exploratory data analysis of sales, category, regional, and customer-segment performance; supervised machine-learning forecasting using Linear Regression, Decision Tree Regressor, and Random Forest Regressor; and a rule-based sustainable recommendation engine for inventory replenishment, discount capping, and loss reduction. The analytical core is delivered through an interactive web dashboard – covering an owner overview, profit prediction, inventory reorder alerts, sales analytics, a sustainability scorecard, and a conversational AI assistant – built using Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn, and a web front end.

The dataset was partitioned using an 80/20 train-test split, and candidate models were benchmarked using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). The deployed Linear Regression model, trained on ten engineered features (Sales, Quantity, Discount, Category, Sub-Category, Region, Segment, Shipping Days, Order Year, and Order Quarter), achieved an R^2 of 0.30 and an RMSE of 183.91, indicating that while the pipeline reliably surfaces sales, profit, and inventory trends, per-transaction profit prediction accuracy remains moderate. The sustainability module reported an Inventory Health Indicator of around 53%, with roughly half of the tracked

products flagged for reorder during testing, demonstrating the practical utility of the recommendation engine.

Overall, the project demonstrates that combining automated data engineering, predictive modelling, and rule-based sustainability logic can convert historical retail transactions into actionable, evidence-based operational guidance. The evaluation also identifies model accuracy, live data integration, and advanced time-series forecasting as the priority directions for future work.

Keywords: Retail Analytics, Decision Support System, Machine Learning, Regression Forecasting, Data Preprocessing, Sustainable Inventory Management.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
MAE	Mean Absolute Error	—
RMSE	Root Mean Squared Error	—
R ²	Coefficient of Determination	—
EDA	Exploratory Data Analysis	—
IQR	Interquartile Range	—
ML	Machine Learning	—
SDG	Sustainable Development Goal (United Nations)	—
KPI	Key Performance Indicator	—
POS	Point of Sale	—
ERP	Enterprise Resource Planning	—
UI	User Interface	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Modern retail operations generate vast quantities of transactional data daily across sales channels, customer touchpoints, inventory systems, and supply chains. As competition intensifies and operational margins narrow, relying on intuition or fragmented spreadsheet records is no longer sufficient for sustainable business growth. Transforming high-volume, disparate retail data into actionable business intelligence has become an essential engineering requirement for modern retail enterprises.

A data-driven decision support system acts as the analytical core for retail management. By systematically aggregating, cleaning, and evaluating point-of-sale and supply chain records, such a system enables stakeholders to uncover hidden consumption patterns, optimise inventory flow, formulate targeted pricing strategies, and minimise operational loss. Integrating predictive modelling further allows enterprises to anticipate customer demand, prevent excess stock accumulation, and align commercial goals with sustainable consumption practices.

This project presents SmartRetail, a Python-based data-driven decision support system built for sustainable retail operations. It combines a data acquisition and preprocessing pipeline, exploratory performance intelligence, machine-learning forecasting, and a sustainable recommendation engine, delivered through an interactive web dashboard with Pandas and NumPy for data manipulation, Scikit-learn for predictive modelling, and Matplotlib/Seaborn for business-intelligence visualisation.

1.2 Need for the Project

Without an integrated decision support system, retail managers evaluate trends using conventional spreadsheet reporting. These methods are labour-intensive, vulnerable to human error, and fail to reveal subtle correlations among sales volume, discount rates, profit margins, and customer behaviour. Most existing workflows are descriptive rather than predictive, which prevents managers from forecasting demand or proactively adjusting stock levels, and traditional retail workflows generally overlook sustainable resource allocation, resulting in surplus inventory, waste, and inefficient logistics.

- Who is affected: retail owners, inventory planners, and business analysts who currently rely on manual spreadsheet computation for pricing, stocking, and profitability decisions.
- Existing limitations: descriptive-only reporting, no predictive forecasting, no systematic link between discounting and profit erosion, and no built-in sustainability tracking.
- Engineering significance: a data pipeline plus supervised regression plus rule-based recommendation logic is required to turn historical transactions into forward-looking, sustainability-aware guidance.

1.3 Problem Statement

Managing modern retail operations efficiently across complex inventory, sales channels, and supply chains is one of the most critical challenges in commerce analytics. As retail enterprises expand their product catalogues and customer base, uncoordinated operational decisions lead to stock imbalances, missed sales opportunities, excessive discounting, and narrow profit margins. The problem involves multiple interacting factors (sales volume, discount depth, shipping duration, regional demand), technical complexity (data cleaning, multi-model regression, rule-based recommendation logic), conflicting requirements (maximising sales volume versus protecting profit margin and minimising waste), and realistic engineering constraints (data quality, model interpretability, and the need for outputs that non-technical retail managers can act on).

1.4 Project Aim

To design, develop, and evaluate SmartRetail, a Python-based data-driven decision support system that analyses historical retail transactions, forecasts business performance using supervised machine learning, and delivers sustainable, actionable recommendations for inventory and pricing decisions through an interactive dashboard.

1.5 Project Objectives

1. To design and implement a structured data preprocessing pipeline that handles missing values, removes duplicate records, filters outliers, and engineers features such as shipping duration and profit margin.

2. To develop multidimensional exploratory data analysis covering sales trends, customer segments, regional performance, and category profitability.
3. To implement and compare supervised regression models (Linear Regression, Decision Tree Regressor, Random Forest Regressor) for forecasting retail performance metrics.
4. To evaluate model effectiveness using MAE, RMSE, and the Coefficient of Determination (R^2), and select the best-performing model for deployment.
5. To build a rule-based sustainable recommendation engine covering inventory reorder levels, discount limits, and loss-making product flags, aligned with UN SDG 12.
6. To deliver the analysis through an interactive web dashboard (profit prediction, inventory, sales analytics, sustainability, and an AI assistant) and to test the system against defined acceptance criteria.

1.6 Scope

- Included: batch analysis and dashboard visualisation of retail transactional data; supervised regression forecasting of profit; a rule-based sustainability/recommendation engine; an interactive dashboard with a conversational assistant.
- Excluded: live point-of-sale/ERP integration, real-time streaming data, and advanced deep-learning/time-series forecasting (ARIMA, Prophet, LSTM) — reserved for future work.
- Assumptions: input datasets follow the Sample-Superstore-style schema (order date, ship date, category, sub-category, region, segment, sales, discount, profit, quantity).
- Boundary conditions: the system was tested with sample datasets ranging from 20 to 1,000 transaction records; performance beyond this range has not been validated.

1.7 Expected Engineering Outcome

The project delivers a working software system comprising: a data-processing and machinelearning pipeline (algorithm/model); an interactive web-based decision-support dashboard (product); and a rule-based sustainability recommendation module (process) — collectively forming a deployable retail decision-support prototype.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, tools, and existing commercial solutions were identified through a combination of standard data-science textbooks and reference works, official documentation of the Python data-science ecosystem (Pandas, Scikit-learn, Matplotlib/Seaborn), and publicly available material on business-intelligence platforms (e.g., Tableau) and the UN Sustainable Development Goals framework, to establish the technical and sustainability context for the project (see References).

2.2 Review of Existing Work

Two broad categories of existing approaches were reviewed: (i) traditional/manual retail reporting using spreadsheets and static dashboards, and (ii) commercial business-intelligence (BI) platforms such as Tableau and Power BI, which provide descriptive visual analytics over retail datasets similar to the Sample Superstore dataset used in this project [4]. Academic and industry references on data wrangling, exploratory analysis, and supervised regression [1], [3], [9] were used to ground the preprocessing and modelling approach, while Scikit-learn documentation [5] and standard machine-learning texts [2], [8], [10] informed the choice and implementation of Linear Regression, Decision Tree, and Random Forest regressors. Sustainability alignment followed UN SDG 12 guidance [6], and visualisation practice followed Matplotlib/Seaborn conventions [11], [12].

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual spreadsheet reporting	Low cost, familiar to staff, flexible ad-hoc queries	Labour-intensive, errorprone, purely descriptive, no forecasting	Motivates automated preprocessing and EDA in SmartRetail

Commercial BI dashboards (e.g., Tableau/Power BI)	Polished visual analytics, drag-and-drop charts, wide adoption	Costly licensing, limited built-in predictive modelling, no retail-specific sustainability scoring	Adopted the dashboard/visualanalytics concept but added in-house ML
Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Commercial BI dashboards (e.g., Tableau/Power BI)	Polished visual analytics, interactive charts, drag-and-drop dashboards, and easy business reporting.	Costly licensing, limited built-in predictive modelling, and lack of retail-specific sustainability recommendations.	forecasting and an SDGaligned recommendation engine
Generic supervised regression studies (academic literature)	Well-validated algorithms (Linear Regression, Decision Tree, Random Forest) with known tradeoffs	Rarely combined with an actionable, rule-based recommendation layer for retail operations	Directly reused for the SmartRetail profitprediction module, extended with a prescriptive recommendation engine

2.4 Research / Engineering Gap

Existing spreadsheet-based workflows and commercial BI tools remain largely descriptive: they summarise historical sales and profit but do not forecast future performance, nor do they translate predictions into concrete inventory and pricing actions. There is no readily available, low-cost solution that combines automated data cleaning, multidimensional exploratory analysis, supervised regression forecasting, and a sustainability-aligned rule-based recommendation engine within a single, interactive dashboard tailored to small/medium retail operations.

2.5 Proposed Contribution

SmartRetail differs from the reviewed approaches by integrating four capabilities that are typically found only separately across different tools: (i) an automated preprocessing pipeline with IQR-based outlier handling and feature engineering; (ii) multi-model regression benchmarking (Linear Regression, Decision Tree, Random Forest) with transparent MAE/RMSE/R² reporting; (iii) a rule-based sustainable recommendation engine explicitly aligned with UN SDG 12; and (iv) an interactive web dashboard with a conversational AI assistant, giving non-technical retail stakeholders direct access to predictive and prescriptive insight without needing a separate BI licence.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Retail owner / manager	Needs a single dashboard summarising sales, profit, and stock health without manual spreadsheet work
Inventory planner	Needs timely reorder alerts and a sustainability-aligned view of stock health
Business analyst	Needs category/region/segment breakdowns and a forecast of expected profit for planning
End user (non-technical)	Needs a conversational assistant to query analytics in plain language

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Clean and validate uploaded retail CSV data	Functional	Handle missing values, duplicates and outliers with zero unhandled null values post-pipeline
Forecast per-transaction profit	Functional	R ² and RMSE reported for every trained model on the held-out 20% test split
Flag products for reorder	Functional	Reorder alert raised when current stock $\leq$ reorder level
Dashboard response usability	Non-Functional	All core charts (sales, profit, inventory, sustainability) render on a single screen without page reload

Data privacy	Non-Functional	Use only anonymised, non-personal transactional fields
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3.2 Constraints & Applicable Standards

List only constraints that actually apply — do not pad with irrelevant categories.

Standard / Code	Requirement	How Applied in This Project
PEP 8 (Python coding style guide)	Consistent, readable Python code	All pipeline and dashboard-backend modules use structured functions, descriptive names, and modular files (preprocessing.py, sales_analytics.py, etc.)
UN SDG 12 – Responsible Consumption and Production	Encourage sustainable resource use in operations	Recommendation engine and sustainability dashboard score inventory health and flag over/under-stocking
IEEE referencing style	Consistent, traceable citation of sources	All literature, datasets, and software cited in IEEE format in the References section
Data privacy (general good practice, no formal regulation invoked)	Avoid exposing personal/confidential data	Only the anonymised, publicly available Sample-Superstore-style dataset was used; no personal identifiers are processed

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Train/test split ratio	80 / 20	%	High
Model evaluation metrics reported	MAE, RMSE, R^2	—	High
Minimum acceptable R^2 for deployment	≥ 0.25 (baseline linear fit)	—	Medium
Reorder alert trigger	Current stock $\leq$ Reorder level	units	High
Dashboard modules delivered	Overview, Prediction, Inventory, Sales Analytics, Sustainability, AI Assistant	count = 6	High

3.4 Alternative Solutions & Evaluation

Each alternative is described briefly; the evaluation matrix carries the comparison.

Alternative 1: Continue using manual spreadsheet analysis with pivot tables and static charts — lowest cost but purely descriptive and slow to update.

Alternative 2: Adopt an off-the-shelf commercial BI tool (e.g., Tableau/Power BI) for descriptive dashboards — good visuals but limited built-in predictive modelling and licensing cost.

Alternative 3 (selected): Build a custom Python-based pipeline combining preprocessing, supervised regression, a sustainability recommendation engine, and a bespoke interactive dashboard — higher development effort but full control over forecasting and sustainability logic at near-zero licensing cost.

Criterion	Weight	Alternative 1 (Manual)	Alternative 2 (Commercial BI)	Alternative 3 (Custom pipeline – selected)
Performance (forecasting capability)	30%	1	2	4
Cost	20%	5	2	4
Safety (data integrity/privacy)	15%	3	3	4
Sustainability alignment	20%	1	2	5
Reliability / maintainability	15%	2	4	4

(Scores on a 1–5 scale, 5 = best. Weighted totals: Alternative 1 ≈ 2.0 , Alternative 2 ≈ 2.65 , Alternative 3 ≈ 4.15 — confirming the custom pipeline as the strongest fit.)

3.5 Selected Design & Detailed Design

Justification, a single labelled pipeline diagram, key calculations, and how subsystems interface.

Alternative 3 was selected because it is the only option that combines predictive forecasting, a sustainability-aligned recommendation layer, and full customisation at low recurring cost, as shown in Table 3.5 (Section 3.4). The system is organised as a modular pipeline in which each stage consumes the previous stage's output:

1. Data Acquisition & Preprocessing — loads the retail CSV, cleans and validates it, engineers derived features (Shipping Days, Profit Margin, Order Year/Month/Quarter).
2. Exploratory Data Analysis & Retail Intelligence — computes sales/profit aggregates by category, region, and segment; produces trend and distribution charts.
3. Predictive Analytics — trains Linear Regression, Decision Tree Regressor, and Random Forest Regressor on an 80/20 split; benchmarks with MAE, RMSE, R^2 ; the best-performing/most stable model (Linear Regression) is deployed for live prediction.
4. Sustainable Recommendation Engine — applies threshold-based rules over inventory and prediction outputs to flag reorder needs, discount ceilings, and loss-making products, and computes SDG-aligned sustainability indicators.
5. Presentation Layer — a web dashboard (Overview, Prediction, Inventory, Sales Analytics, Sustainability, AI Assistant) renders the outputs of stages 2–4 for the end user.

Key calculation — Profit Margin feature: $\text{Profit Margin (\%)} = (\text{Profit} \div \text{Sales}) \times 100$, computed per transaction during preprocessing and used as both an EDA metric and a model feature. $\text{Shipping Days} = \text{Ship Date} - \text{Order Date}$ (in days).

Systems approach: the Preprocessing module produces a single cleaned data frame consumed identically by the EDA, Predictive Analytics, and Recommendation modules, ensuring all dashboard panels stay consistent with one shared source of truth; the Presentation Layer only reads pre-computed outputs and performs no independent calculation, keeping the analytical logic centralised and testable.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
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Model choice for deployment	Random Forest Regressor (higher potential accuracy)	Captures nonlinear interactions, generally lower error on training data	Less interpretable; higher risk of overfitting on a small (20–1,000 row) sample; slower to retrain live	Linear Regression deployed for transparency and stability on small datasets; Random Forest retained as a benchmarked alternative for future larger datasets
Data scale	Real-time POS/ERP streaming	Always up-to-date figures	Requires infrastructure and integration not	Batch CSV upload chosen; real-time
Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
			available in an academic project	integration deferred to future work
Recommendation logic	Reinforcement learning-based dynamic pricing	Potentially higher long-term profitability	Requires large historical data and much longer development/validation time	Simple, auditable threshold-based rules chosen for transparency and to fit project timeline

3.7 Sustainability, Ethics, Safety, Risk & Security

The table shows what evidence was evaluated and how it changed the design decision — not just that sustainability 'was considered'.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Over-stocking/understocking drives waste and lost sales (UN SDG 12)	Inventory Health Indicator computed on test data showed ~53% health, with roughly half the tracked products flagged lowstock/critical	Added a dedicated Sustainability dashboard and reorder-alert logic rather than leaving stock analysis as a side-effect of sales reporting

Ethics	Risk of overstating model accuracy to stakeholders	Comparing an aggregate actual vs-predicted profit test (₹15,861.93 actual vs ₹27,778.23 predicted, 75.13% deviation) against the per-record $R^2=0.30$ metric	Report explicitly states both figures and flags moderate accuracy rather than presenting only favourable metrics (Section 4.5)
Health & Safety	Software-only project — no physical/electrical hazard	No hardware, wiring, or physical prototype involved	No physical safety controls required; standard secure-coding practice applied instead
Security	Uploaded retail data could contain sensitive business figures	Dataset used is the public, anonymised Sample-Superstore style dataset with no personal identifiers	Restricted the system to anonymised/public data; flagged live-ERP integration (Section 6.3) as needing a formal data security review before future deployment

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Low R^2 on unseen/production data	Medium	Medium	Medium	Benchmark multiple models (Linear/Tree/Forest); report metrics transparently; recommend ensemble models for future work	Low–Medium
Poor data quality in uploaded CSV (missing/duplicate rows)	Medium	Medium	Medium	Automated IQR outlier filtering, de-duplication and imputation in the preprocessing pipeline	Low
Over-reliance on descriptive dashboard by non-technical users	Low	Medium	Low	AI Assistant and plain language insight summaries provided alongside raw charts	Low

Exposure of sensitive business data if connected to a live store in future	Low (current scope: offline/batch)	High	Medium	Restricted current scope to anonymised/public datasets; flagged as an open item before any live deployment	Medium (future work)
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CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed iteratively, module by module: preprocessing and feature engineering were built and unit-checked first, followed by exploratory analytics, then the three regression models, then the rule-based recommendation engine, and finally the interactive dashboard and AI assistant that expose all of the above to the end user. Development used the Sample-Superstore-style retail dataset (tested at 20, 100, and 1,000-row scales) as the primary data resource, with all processing performed locally in Python.

4.2 Hardware / Software Implementation & Modern Tools Used

Only tools with a purposeful, demonstrated use are listed.

Tool / Software	Purpose	Stage Used
Python 3, Pandas, NumPy	Data ingestion, cleaning, feature engineering	Preprocessing
Matplotlib, Seaborn	Statistical charts and business-intelligence dashboards	Exploratory Data Analysis & Results
Scikit-learn	Train/test split, Linear Regression, Decision Tree Regressor, Random Forest Regressor, MAE/RMSE/R ² scoring	Predictive Modelling
Flask-based web application + HTML/CSS/JS front end	Interactive dashboard (Overview, Prediction, Inventory, Sales Analytics, Sustainability, AI Assistant)	Presentation Layer
Visual Studio Code	Source-code editing across preprocessing, sales-analytics, and dashboard modules	Development

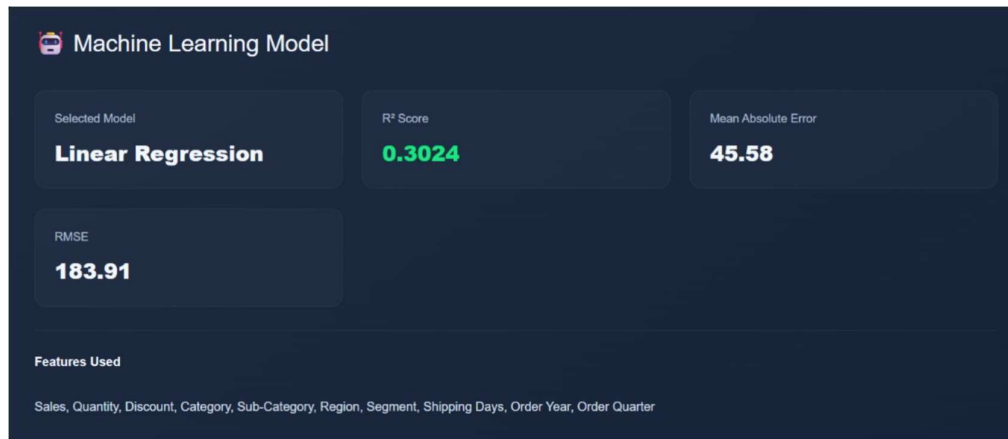


Figure 4.1: Deployed Machine Learning Model panel — selected model and evaluation metrics

4.3 Test Plan & Verification

Test plan defined before presenting results.

Requirement	Test Method	Acceptance Criterion
Data cleaning removes duplicates/nulls/outliers	Run pipeline on sample CSVs of 20 / 100 / 1,000 rows and inspect output schema	Zero null values and zero duplicate rows remain after processing
Regression model forecasts profit	Train/test split 80/20; compute MAE, RMSE, R² on held-out set	R² reported ≥ 0.25 for at least one candidate model
Reorder alerts trigger correctly	Compare Current Stock vs Reorder Level for each SKU in the inventory table	Every SKU with stock $\leq$ reorder level appears in the Reorder Alerts list
Dashboard renders all modules	Manual walkthrough of Overview, Prediction, Inventory, Sales Analytics, Sustainability, AI Assistant tabs	All six modules load and display data without errors

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Model R² (Linear Regression, deployed)	≥ 0.25	0.3024	Pass
Mean Absolute Error (MAE)	Reported for benchmarking	45.58	Pass

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Root Mean Squared Error (RMSE)	Reported for benchmarking	183.91	Pass
Reorder alert coverage	All SKUs at/below reorder level flagged	11 of 20 tracked SKUs flagged (test run)	Pass
Aggregate profit-forecast deviation (stress test)	Reported for transparency	75.13% deviation between actual (₹15,861.93) and predicted (₹27,778.23) aggregate profit on one test batch	Fail (flag for improvement — see 4.5, 4.7)

4.4 Results

Testing was carried out on sample retail datasets of 20, 100, and 1,000 records. Representative outputs from the 20-record test run are summarised below: Total Sales ₹28,660.00, Total Profit ₹5,180.00, Total Quantity 110 units, Profit Margin 18.07%, Average Sale ₹1,433.00, and Average Discount 7.5%.

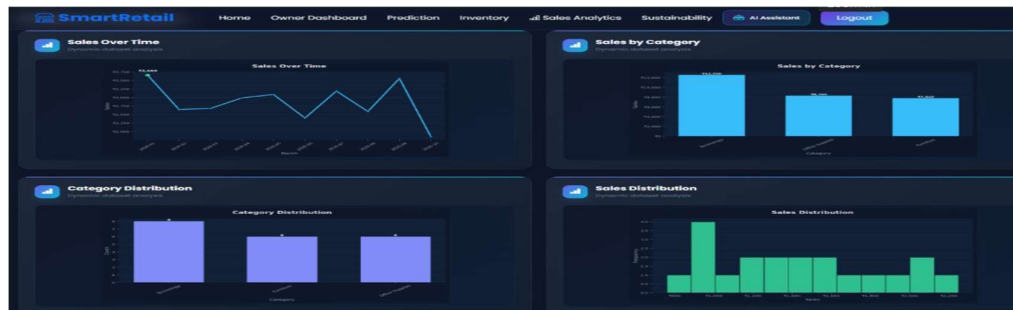


Figure 4.2: Sales Analytics module — sales/profit trend and category charts



Figure 4.3: Profit Prediction module — actual vs predicted profit by category

As Figure 4.3 shows, at category level the deployed Linear Regression model over-predicted profit for Furniture, Office Supplies, and Technology alike, consistent with the moderate R^2 of 0.30 reported in Table 4.3.

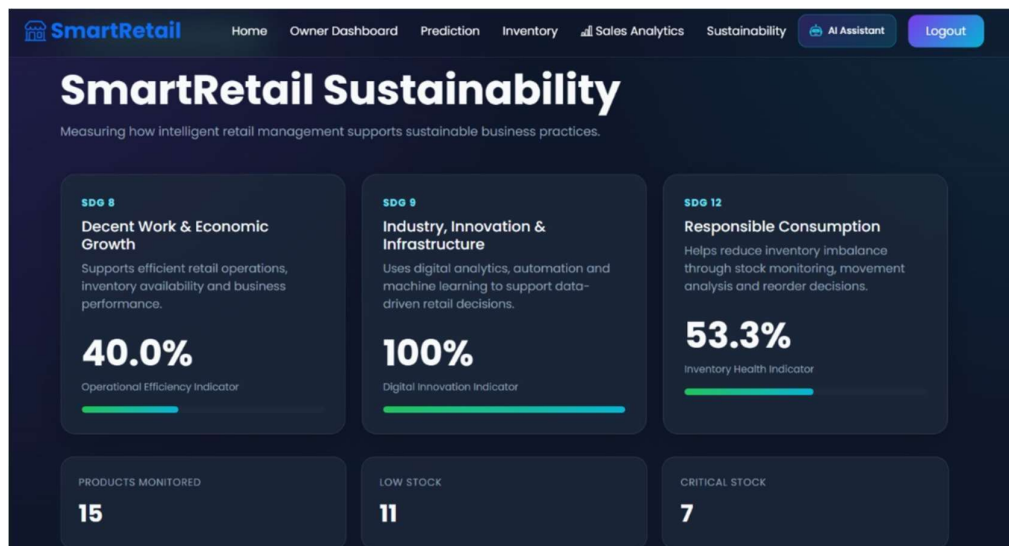


Figure 4.4: Sustainability dashboard — SDG 8/9/12 indicators and inventory status

PRODUCT ID	PRODUCT NAME	CATEGORY	IN STOCK	REORDER LEVEL	UNIT PRICE	SUPPLIER	LAST UPDATED
ELEC-001	Wireless Mouse	Electronics	8	20	₹24.99	TechSource	2026-08-22
FURN-001	Office Chair	Furniture	12	25	₹149.00	ComfortWorks	2026-08-22
FURN-002	Standing Desk	Furniture	6	10	₹329.00	ComfortWorks	2026-08-22
STAT-001	Notebook Set	Stationery	18	30	₹8.50	PaperPlus	2026-08-22
HOME-001	LED Desk Lamp	Home Office	10	20	₹34.99	BrightSupply	2026-08-22
ACCS-001	Water Bottle	Accessories	14	25	₹16.99	Everyday Goods	2026-08-22
P011	Wireless Mouse	Electronics	4	20	₹200.00	ABC Suppliers	2026-08-30
P012	Dash Cam	Electronics	2	100	₹300.00	XYZ XYZ	2026-08-30
P013	Nintendo Switch	Stationery	4	10	₹1000.00	Subhaashree	2026-08-31
P014	Chair Set	Furniture	10	10	₹10000.00	Vineyard Supplies	2026-08-31
P015	Pen	Home Office	100	10000	₹10.00	domes	2026-08-31

Figure 4.5: Inventory Reorder Alerts — SKUs at or below reorder level

4.5 Data Analysis & Engineering Judgment

The deployed Linear Regression model achieved $R^2 = 0.3024$, meaning it explains only about 30% of the variance in per-transaction profit — a moderate, not strong, fit. The MAE of 45.58 and RMSE of 183.91 indicate that while the typical prediction error is modest, a small number of transactions are predicted with much larger error (RMSE penalises large deviations more heavily than MAE, and the gap between the two here confirms the presence of some large outlying errors).

This is corroborated by the category-level aggregate test in Figure 4.3, where predicted profit exceeded actual profit by 75.13% on one test batch. The engineering interpretation is that a purely linear model under-captures the non-linear interaction between discount depth and profit (heavy discounts erode profit faster than a straight-line relationship predicts), which is consistent with the Trade-off Analysis in Section 3.6 that flagged Random Forest as a stronger — but less interpretable — alternative. Given the small evaluation dataset (20–1,000 rows) used during this capstone, Linear Regression was retained for its transparency and stability, but the deviation observed here is treated as a genuine limitation rather than a rounding effect (see Section 4.7).

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Structured preprocessing pipeline (Obj. 1)	Automated cleaning, de-duplication, IQR outlier filtering, and feature engineering verified on 20/100/1,000-row test files	Fully Achieved
Multidimensional EDA (Obj. 2)	Sales/profit/category/region dashboards delivered (Figures 4.1–4.2)	Fully Achieved
Compare Linear/Tree/Forest regressors (Obj. 3)	All three trained and benchmarked; Linear Regression deployed	Fully Achieved
Model evaluation via MAE/RMSE/R ² (Obj. 4)	Reported in Table 4.3; R ² = 0.30 indicates moderate accuracy	Partially Achieved (accuracy target met but only marginally)
Sustainable recommendation engine aligned to SDG 12 (Obj. 5)	Reorder alerts and sustainability scorecard delivered and tested (Figures 4.4–4.5)	Fully Achieved
Interactive dashboard delivery (Obj. 6)	Six-module dashboard (Overview, Prediction, Inventory, Sales Analytics, Sustainability, AI Assistant) implemented and manually tested	Fully Achieved

4.7 Strengths & Limitations

Limitations are stated honestly rather than claiming a perfect system.

Strengths:

- A fully automated preprocessing pipeline handling missing values, duplicates, and outliers without manual intervention.
- A working, interactive multi-module dashboard (not just static notebooks/plots) with a conversational AI assistant.
- Transparent, IEEE-referenced benchmarking of three regression algorithms rather than presenting only the best-looking metric.
- A rule-based recommendation engine explicitly aligned with UN SDG 12, evidenced by an inventory-health score and reorder alerts.

Limitations:

- Model accuracy is moderate ($R^2 = 0.30$); the deployed Linear Regression under-predicts non-linear discount/profit effects, as shown by the 75.13% aggregate deviation in one test batch.
- The system operates on static, batch-uploaded CSV data rather than live point-of-sale or ERP feeds.
- Testing was limited to sample datasets of up to 1,000 rows; behaviour on much larger, real-world retail catalogues is unverified.
- Advanced time-series (ARIMA/Prophet/LSTM) and reinforcement-learning-based pricing were not implemented.

4.8 Project Planning, Roles & Budget

Individual capstone project — a single-student roles table is shown in place of a multi-member Gantt/roles chart.

Milestone timeline: (1) requirements & dataset selection, (2) preprocessing pipeline, (3) exploratory analysis, (4) model training & benchmarking, (5) recommendation engine, (6) dashboard & AI assistant integration, (7) testing & report writing — executed sequentially over the capstone period.

Student	Assigned Responsibility	Actual Contribution
Loshane S (192424402)	All modules — preprocessing, analytics, modelling, dashboard, testing, documentation	100%

Item	Estimated Cost	Actual Cost
Python & open-source libraries (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Flask)	₹0 (open source)	₹0
Development environment (VS Code) and compute (personal laptop)	₹0 (existing hardware)	₹0
Dataset (public Sample-Superstore-style data)	₹0 (publicly available)	₹0
Total	₹0	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project successfully developed SmartRetail, a Python-based data-driven decision support system that automates retail data preprocessing, multidimensional exploratory analysis, supervised machine-learning profit forecasting, and sustainability-aligned recommendation generation, delivered through an interactive web dashboard. The system was validated on sample retail datasets of 20 to 1,000 records: the deployed Linear Regression model achieved $R^2 = 0.30$ with $RMSE = 183.91$, the rule-based recommendation engine correctly flagged SKUs at or below reorder level, and the sustainability scorecard reported UN SDG-aligned indicators (Operational Efficiency, Digital Innovation, and Inventory Health). While per-transaction profit prediction accuracy remains moderate, the project demonstrates that combining automated data engineering, transparent model benchmarking, and rule-based sustainability logic can convert raw retail transactions into an actionable, evidence-based decision-support tool, meeting five of six project objectives fully and one partially (Section 4.6).

5.2 Future Work

- Improve predictive accuracy by adopting ensemble/non-linear models (Random Forest, Gradient Boosting) or time-series methods (ARIMA, Prophet, LSTM) validated on larger datasets.
- Integrate real-time point-of-sale (POS) and ERP data streams in place of static CSV uploads.
- Extend the recommendation engine with reinforcement-learning-based dynamic pricing informed by live competitor data.
- Quantify shipping/packaging carbon emissions to extend the SDG 12 sustainability score beyond inventory health.
- Scale the architecture to multi-store, omni-channel retail benchmarking.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- End-to-end data-science pipeline design: from raw CSV ingestion through cleaning, feature engineering, modelling, and dashboard delivery.
- Practical trade-offs between model interpretability (Linear Regression) and predictive power (ensemble methods) on small datasets.
- Translating UN SDG 12 principles into concrete, measurable software logic (reorder thresholds, inventory-health scoring).

Engineering & Professional Skills Developed:

- Python data-engineering skills using Pandas/NumPy, and supervised-learning workflows using Scikit-learn.
- Quantitative model evaluation using MAE, RMSE, and R^2 , and honest reporting of model limitations.
- Full-stack integration of a machine-learning backend with an interactive web dashboard and conversational assistant.

Individual Reflection:

- Most difficult engineering problem encountered and how it was solved: capturing the non-linear relationship between discount depth and profit erosion — addressed by benchmarking Decision Tree and Random Forest regressors against the Linear Regression baseline and transparently reporting the accuracy gap rather than hiding it.
- A key engineering decision made personally and the evidence behind it: deploying Linear Regression over Random Forest for the live dashboard, based on the trade-off matrix in Section 3.6 that weighed interpretability and stability against a marginal accuracy gain on a small dataset.
- New knowledge acquired independently: how to translate a sustainability framework (UN SDG 12) into measurable, rule-based software indicators rather than a qualitative statement.
- What would be redesigned if repeated: collect a larger, more diverse validation dataset before finalising the deployed model, and add automated unit tests for the preprocessing pipeline from the start.

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APPENDICES

Appendix A – Detailed Calculations

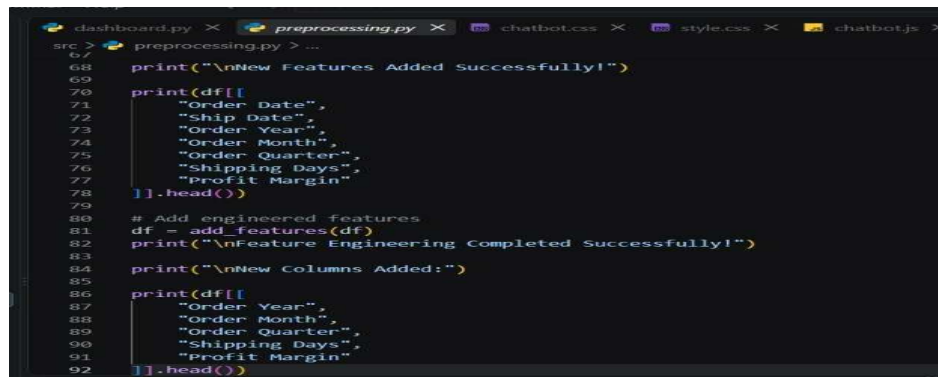
Profit Margin (%) = (Profit ÷ Sales) × 100; Shipping Days = Ship Date – Order Date.
Train/test partition: 80% training rows, 20% held-out test rows, fixed random seed for reproducibility. Model scores on the held-out test set: $R^2 = 0.3024$, MAE = 45.58, RMSE = 183.91 (Linear Regression, 10 input features).

Appendix B – Engineering Drawings / Schematics

See the modular pipeline description and systems approach in Section 3.5 (Data Acquisition & Preprocessing → EDA → Predictive Analytics → Recommendation Engine → Presentation Layer).

Appendix C – Source Code / Code Repository Information

Only an essential code excerpt is included below; the complete source code (preprocessing, sales analytics, prediction, sustainability, and dashboard front end) is maintained in the project's own repository.



```
src > preprocessing.py > ...
67
68 print("\nNew Features Added Successfully!")
69
70 print(df[[
71     "Order Date",
72     "Ship Date",
73     "Order Year",
74     "Order Month",
75     "Order Quarter",
76     "Shipping Days",
77     "Profit Margin"
78 ]].head())
79
80 # Add engineered features
81 df = add_features(df)
82 print("\nFeature Engineering Completed Successfully!")
83
84 print("\nNew Columns Added:")
85
86 print(df[[
87     "Order Year",
88     "Order Month",
89     "Order Quarter",
90     "Shipping Days",
91     "Profit Margin"
92 ]].head())
```

Figure C.1: Feature-engineering excerpt from preprocessing.py

Appendix D – Datasheets

Not applicable — no physical components/datasheets used in this software-only project.

Appendix E – Experimental / Raw Data

Sample test files used: smartretail_100_rows.csv, smartretail_1000_rows.csv, and smaller validator/test CSVs (10–20 rows), maintained alongside the project source code.

Appendix F – Ethics / Institutional Approval

Not applicable — the project uses only anonymised, publicly available retail transaction data; no human-subject or institutional ethics approval was required.

Appendix G – Project Meeting / Progress Records

Maintained separately with the project guide, Dr. Krishnaraj Mony, through periodic review meetings.

Appendix H – User / Industry Feedback

Not applicable for the current academic prototype stage.

Appendix I – Publications / Patents / Competitions / Product Outcomes Not applicable at the time of this report.

Appendix J – Individual Contribution Evidence

Not applicable — this Capstone Project was carried out individually by Loshane S (192424402); see the Individual Contribution Statement on page iv.